

Practical-3

Practice lab Assignment

The screenshot displays the CodeTANTRA web interface for a Python assignment. The left sidebar contains the problem description and test cases. The main editor shows the solution code, and the right sidebar shows the test results.

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching**: Find the indices where `search_value` appears in `array1` and print these indices.
- Counting**: Count how many times `count_value` appears in `array1` and print the count.
- Broadcasting**: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- Sorting**: Sort `array1` in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing `array1`.
- An integer `search_value` represents the value to search for in the array.
- An integer `count_value` represents the value to count in the array.
- An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

- The indices where `search_value` occurs in `array1`.
- The count of occurrences of `count_value` in `array1`.
- The array after adding the `broadcast_value` to each element.
- The sorted array.

Sample Test Cases

arrayOpe...

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 search_indices = np.where(array1 == search_value)[0]
13 print(search_indices)
14
15 # Count occurrences in array1
```

Average time: 0.024 s, Maximum time: 0.030 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed.

Test case 1

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3

Terminal, Test cases

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Lab Assignment

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3.2.5. Numpy: Copying and Viewing Arrays01:45

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>
View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

1import numpy as np
2
3inputlist = list(map(int,input().split(" ")))
4
5# Original array
6original_array = np.array(inputlist)
7
8# Create a view
9view_array =original_array.view()
10
11# Create a copy
12copy_array =original_array.copy()
13
14# Modify the view
15view_array[0] = 99

Average time0.016 s
16.25 ms

Maximum time0.020 s
20.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 120 ms

Expected output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]

Actual output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]

Terminal

Test cases

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays12:40

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

1import numpy as np
2
3# Input matrices
4print("Enter Array1:")
5arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7print("Enter Array2:")
8arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10# Perform horizontal stacking (hstack)
11a=np.hstack((arr1,arr2))
12print("Horizontal Stack:")
13print(a)
14
15

Average time0.052 s
62.00 ms

Maximum time0.088 s
88.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 168 ms

Expected output
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6

Actual output
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6

Terminal

Test cases

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

The program should print the generated sequence based on the input values.

Sample Test Cases

customs...

```
1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate the sequence using np.arange()
9 a=np.arange(start,stop,step)
10 print(a)
11
12 # Print the generated sequence
13
14
```

Average time0.023 s22.75 ms

Maximum time0.025 s25.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 125 ms

Expected output

Actual output

3	3
10	10
2	2
[3 5 7 9]	[3 5 7 9]

Test case 225 ms

TerminalTest cases

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operations

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_1 OR B_1).

Input Format:

The first line contains space-separated integers representing the elements of array A.

The second line contains space-separated integers representing the elements of array B.

Output Format:

For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     A=np.array(A)
7     B=np.array(B)
8
9     # Arithmetic Operations
10    sum_result = A+B
11    diff_result = A-B
12    prod_result = A*B
13
14    # Statistical Operations
15    mean_A = np.mean(A)
```

Average time0.024 s24.33 ms

Maximum time0.043 s43.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 143 ms

Expected output

Actual output

1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5

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3.2.5. Numpy: Copying and Viewing Arrays

01:45

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: `<original_array>`
View array: `<view_array>`

- After modifying the copy:

Original array after modifying copy: `<original_array>`
Copy array: `<copy_array>`

Sample Test Cases

copyAnd...

Submit

Debugger

1import numpy as np

2

3inputlist = list(map(int,input().split(" ")))

4

5# Original array

6original_array = np.array(inputlist)

7

8# Create a view

9view_array =original_array.view()

10

11# Create a copy

12copy_array =original_array.copy()

13

14# Modify the view

15view_array[0] = 99

Average time0.016 s16.25 ms

Maximum time0.020 s20.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 120 msDebug

Expected output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Actual output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Terminal

Test cases

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